

SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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1. INTRODUCTION

1.1. Background and motivation.

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$ and let G be the corresponding finite reflection group. Let $k : R \rightarrow \mathbb{R}_+$ be G -invariant multiplicity function. Let $R_{\text{Dunkl},j}$ be the j -th Dunkl-Riesz transform corresponding to R and k . Let w be the Dunkl weight corresponding to R and k . Let $f \in L_1^{\text{loc}}(\mathbb{R}^N)$.*

- (i) *If $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p > N$, then $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,*

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{p,R,k} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

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- (ii) If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, $p > N$, then $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{p,R,k}^{-1} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

- (iii) If $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{R,k} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (iv) If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_{R,k}^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

1.2. Strategy of the proof. In Section 3, we introduce our *ad hoc* Sobolev space and demonstrate that its semi-norm is controlled by the usual Sobolev semi-norm. In Section 4, we use Janson-Wolff method to control the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$ by the *ad hoc* Sobolev semi-norm of f . Thus, we control the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$ by the usual Sobolev semi-norm (i.e., we prove Theorem 1.1 (i)).

In Section 5, we control the oscillation semi-norm of f (with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$) via $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$. After that, we verify the assumptions in Proposition 1.2 in [6] for measures $w(x)dx$ and dx and, using the latter proposition, conclude that oscillation semi-norm of f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$ is equivalent to the one with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$. The latter is known to be equivalent to the usual Sobolev semi-norm of f .

1.3. Notation. Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N , p , and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda \ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p , which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ -measurable set E by

$$(1.1) \quad (f)_E := \int_E f(x) d\mu(x) := \frac{1}{\mu(E)} \int_E f(x) d\mu(x).$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a

nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $|v| = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $k : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper.

Define the Dunkl measure $w(x)dx$ on $\mathbb{R}^N$ by

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)}.$$

Let $B(x, r) := \{y \in \mathbb{R}^N : |x - y| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Furthermore,

$$(2.1) \quad w(B(x, r)) \sim r^N \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on N, R and k . It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2.2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{N'}, \text{ for } 0 < r_1 < r_2,$$

so the numbers N' and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl}, j} := T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$, $j = 1, 2, \dots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$(2.3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define

$$d(x, y) = \min_{\sigma \in G} |x - \sigma(y)|, \quad x, y \in \mathbb{R}^N.$$

Recall from [5, Chapter VII, proof of Theorem 2.12] that

$$(2.4) \quad d(x, y) = |x - y| \text{ if and only if } x, y \in \Gamma \text{ for some chamber } \Gamma.$$

Denote by $K_{\text{Dunkl}, j}(x, y)$ the integral kernel of $R_{\text{Dunkl}, j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [4]. Although

these estimates may not be sharp (see the comment following Theorem 2.1 in [4, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j \in \{1, 2, \dots, N\}$ and every x, y with $d(x, y) \neq 0$,*

$$|K_{\text{Dunkl},j}(x, y)| \leq \frac{d(x, y)}{|x - y|} \cdot \frac{C}{w(B(x, d(x, y)))}.$$

Proof. The proof is given in [4, Theorem 1.1]. $\square$

Lemma 2.3. *We have*

$$d(x, y)^2 \cdot w(x)w(y) \leq Cw^2(B(x, d(x, y))), \quad x, y \in \mathbb{R}^N.$$

Proof. Define equivalence relation $\sim$ on R by saying $v_1 \sim v_2$ if there exists $g \in G$ with $v_1 = gv_2$. Let us split R into equivalence classes with respect to $\sim$. Since k is G -invariant, it follows that k is constant on each equivalence class. Note that G leaves each equivalence class invariant.

We, therefore, write

$$w(x) = \prod_{A \in R/\sim} \left(\prod_{v \in A} |\langle x, v \rangle| \right)^{k|A|}.$$

On the other hand,

$$w(B(x, r)) \geq c_{R,k}^{(1)} r^N \prod_{A \in R/\sim} \left(\prod_{v \in A} (r + |\langle x, v \rangle|) \right)^{k|A|}.$$

Let $x \in \Gamma$ and $y \in g\Gamma$. Set $z = g^{-1}y \in \Gamma$. Note that $d(x, y) = |x - z|$ and, therefore,

$$|\langle x, v \rangle| + d(x, y) = |\langle x, v \rangle| + |x - z| \geq 2^{-\frac{1}{2}} |\langle z, v \rangle|.$$

Thus,

$$\begin{aligned} \prod_{v \in A} (|\langle x, v \rangle| + d(x, y)) &\geq \prod_{v \in A} 2^{-\frac{1}{2}} |\langle z, v \rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle g^{-1}y, v \rangle| = \\ &= 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, gv \rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in gA} |\langle y, v \rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v \rangle|. \end{aligned}$$

Thus,

$$\begin{aligned} w(B(x, d(x, y))) &\geq c_{R,k}^{(1)} d^N(x, y) \prod_{A \in R/\sim} \left(\prod_{v \in A} (d(x, y) + |\langle x, v \rangle|) \right)^{k|A|} \geq \\ &\geq c_{R,k}^{(1)} d^N(x, y) \prod_{A \in R/\sim} \left(2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v \rangle| \right)^{k|A|} = c_{R,k}^{(2)} d^N(x, y) w(y). \end{aligned}$$

On the other hand,

$$w(B(x, d(x, y))) \geq c_{R,k}^{(1)} d^N(x, y) w(x).$$

Multiplying the last two equations, we complete the proof. $\square$

Lemma 2.4. *Let $1 \leq j \leq N$. If $x_j \geq y_j$, then*

$$K_{\text{Dunkl},j}(x, y) \geq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If $x_j \leq y_j$, then

$$K_{\text{Dunkl},j}(x, y) \leq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

Proof. The lower bound is identical to that in [4, Theorem 1.2]. $\square$

2.2. Schatten class. [Zhijie will add this later]

2.3. Nearly weakly orthonormal sequences. Rochberg and Semmes [11] introduced the notion of *nearly weakly orthonormal* (NWO) sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [3, 7, 10, 1, 2, 9, 11, 13, 12]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [7, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Definition 2.5. Let $\mathcal{D}$ be the dyadic lattice in $\mathbb{R}^N$. A sequence of functions $\{e_Q\}_{Q \in \mathcal{D}}$ is called *nearly weakly orthonormal* (NWO, in short) sequence if

- (i) $\text{supp}(e_Q) \subset cQ$ for some positive constant c independent of Q .
- (ii) $\|e_Q\|_{L_\infty(\mathbb{R}^N, w(x)dx)} \leq w(Q)^{-\frac{1}{2}}$.

Lemma 2.6. Let $1 < p < \infty$ and let $\mathcal{D}$ be the dyadic lattice in $\mathbb{R}^N$. If $T \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, then

$$\left\| \left\{ \langle Te_Q, f_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q \in \mathcal{D}} \right\|_{l_{p,q}(\mathcal{D})} \leq c_{N,p,q,w,c_1,c_2} \|T\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N, w(x)dx))}$$

for every NWO systems $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ (with respective constants c_1 and c_2).

Proof. reference to [11] is inappropriate. Rochberg and Semmes do not consider weighted setting. we absolutely must provide correct reference. This property can be found in [11, (1.10)] (see also [7, Section 7]). $\square$

3. COMPARISON OF SOBOLEV NORMS

Theorem 3.1. For every $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p > N$, we have

$$\left(\int_{\mathbb{R}^N \times \mathbb{R}^N} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq c_{p,R} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Lemma 3.2. For every convex polyhedral cone Γ , we have

$$\int_{\Gamma} \frac{dx}{|x-y|^{2N}} \geq c_{\Gamma} \text{dist}(y, \Gamma)^{-N}, \quad y \notin \Gamma.$$

Proof. Find a point $z \in \Gamma$ such that

$$|y - z| = \text{dist}(y, \Gamma).$$

We have $z + \Gamma \subset \Gamma$. Thus,

$$\int_{\Gamma} \frac{dx}{|x-y|^{2N}} \geq \int_{z+\Gamma} \frac{dx}{|x-y|^{2N}} = \int_{\Gamma} \frac{dx}{|x-(y-z)|^{2N}}.$$

Clearly,

$$|x - (y - z)| \leq |x| + |y - z|.$$

Thus,

$$\int_{\Gamma} \frac{dx}{|x - y|^{2N}} \geq \int_{\Gamma} \frac{dx}{(|x| + |y - z|)^{2N}} = |y - z|^{-N} \int_{\Gamma} \frac{dx}{(|x| + 1)^{2N}}.$$

□

Lemma 3.3. *If $p \in (N, 2N)$, then*

$$\sup_{a>0} \int_{\mathbb{R}^N} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} < \infty.$$

Proof. Step 1: we claim that, for every $a > 0$,

$$\begin{aligned} & \int_{\mathbb{R}^N} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} \leq \\ & \leq 4^p \int_{\mathbb{R}^{N-1} \times (-a, a)} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p} + 3^{2N-p} \int_{\mathbb{R}^N} \frac{|x|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

We clearly have

$$\begin{aligned} & \int_{\mathbb{R}^N} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} = \\ & = \int_{\mathbb{R}^{N-1} \times (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} + \int_{\mathbb{R}^{N-1} \times \mathbb{R} \setminus (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

If $x_N \notin (\frac{a}{2}, \frac{3a}{2})$, then $|x_N - a| \geq \frac{1}{3}|x_N|$. Hence, $|x - ae_N| \geq \frac{1}{3}|x|$. Thus,

$$\begin{aligned} & \int_{\mathbb{R}^{N-1} \times \mathbb{R} \setminus (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} \leq \\ & \leq \int_{\mathbb{R}^{N-1} \times \mathbb{R} \setminus (\frac{a}{2}, \frac{3a}{2})} \frac{3^{2N-p} |x|^{p-2N} dx}{(1 + |x|)^p} \leq \int_{\mathbb{R}^N} \frac{3^{2N-p} |x|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

If $x_N \in (\frac{a}{2}, \frac{3a}{2})$, then (here, we using the notation $x' = (x_1, \dots, x_{N-1})$)

$$1 + |x| \geq 1 + \max\{|x'|, \frac{1}{2}a\} \geq \frac{1}{4}(1 + |x'| + a).$$

Thus,

$$\begin{aligned} & \int_{\mathbb{R}^{N-1} \times (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} \leq 4^p \int_{\mathbb{R}^{N-1} \times (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x'| + a)^p} = \\ & = 4^p \int_{\mathbb{R}^{N-1} \times (-\frac{a}{2}, \frac{a}{2})} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p} \leq 4^p \int_{\mathbb{R}^{N-1} \times (-a, a)} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p}. \end{aligned}$$

Combining all preceding estimates, we infer the claim in Step 1.

Step 2: we claim that

$$\sup_{a>0} \int_{\mathbb{R}^{N-1} \times (-a, a)} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p} < \infty.$$

Passing to polar coordinates in $\mathbb{R}^{N-1}$, we equivalently rewrite the claim as

$$\sup_{a>0} \int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1 + r + a)^p} < \infty.$$

If $0 < a < 1$, then

$$\int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(1 + r + a)^p} \leq \int_{\mathbb{R}_+ \times (0, 1)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(1 + r)^p}.$$

If $a \geq 1$, then

$$\begin{aligned} \int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(1 + r + a)^p} &\leq \int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(r + a)^p} = \\ &= \int_{(a, \infty) \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(r + a)^p} + \\ &+ \int_0^a \int_0^r \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dx_N dr}{(r + a)^p} + \int_0^a \int_0^{x_N} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(r + a)^p} \leq \\ &\leq \int_{(a, \infty) \times (0, a)} \frac{r^{p-2N} r^{N-2} dr dx_N}{r^p} + \\ &+ \int_0^a \int_0^r \frac{r^{p-2N} r^{N-2} dx_N dr}{a^p} + \int_0^a \int_0^{x_N} \frac{x_N^{p-2N} r^{N-2} dr dx_N}{a^p} = \\ &= \int_a^\infty r^{-N-1} dr + a^{-p} \int_0^a r^{p-N-1} dr + \frac{1}{N-1} a^{-p} \int_0^a x_N^{p-N-1} dx_N = \\ &= \frac{1}{N} a^{-N} + \frac{1}{p-N} a^{-N} + \frac{1}{(N-1)(p-N)} a^{-N} \leq \frac{1}{N} + \frac{1}{p-N} + \frac{1}{(N-1)(p-N)}. \end{aligned}$$

This yields the claim in Step 2.

Combining Steps 1 and 2, we complete the proof of the lemma. $\square$

Lemma 3.4. *If $p \in (N, 2N)$, then*

$$\int_{\mathbb{R}^{N-1} \times \mathbb{R}_+} \frac{|x - y|^{p-2N} dx}{|x + z|^p} \leq c_{p,N} z_N^{-N}, \quad y, z \in \mathbb{R}^{N-1} \times \mathbb{R}_+.$$

Proof. By dilation invariance, we may assume without loss of generality that $z_N = 1$. By translation invariance in $\mathbb{R}^{N-1}$, we may assume without loss of generality that $z_1 = \dots = z_{N-1} = 0$. We, therefore, have

$$|x + z| = |x + e_N| \geq \max\{1, |x|\} \geq \frac{1}{2}(1 + |x|), \quad x \in \mathbb{R}^{N-1} \times \mathbb{R}_+.$$

Thus,

$$\begin{aligned} \int_{\mathbb{R}^{N-1} \times \mathbb{R}_+} \frac{|x - y|^{p-2N} dx}{|x + z|^p} &\leq 2^p \int_{\mathbb{R}^{N-1} \times \mathbb{R}_+} \frac{|x - y|^{p-2N} dx}{(1 + |x|)^p} \leq \\ &= 2^p \int_{\mathbb{R}^N} \frac{|x - y|^{p-2N} dx}{(1 + |x|)^p} = 2^p \int_{\mathbb{R}^N} \frac{|x - |y|e_N|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

The assertion follows from Lemma 3.3. $\square$

We need the following elementary version of the Hahn–Banach theorem.

Lemma 3.5. *Let $K \subset \mathbb{R}^N$ be a convex set and let $y \notin K$. Let $z \in K$ be the point closest to y . Affine hyperplane P passing through z and orthogonal to $y - z$ splits the space into two half-spaces, one of them containing y , the other one containing K .*

¹It is exactly at this point we are using the assumption $p < 2N$.

Proof. By translation invariance, we may assume without loss of generality that $z = 0$. By rotation and dilation invariance, we may assume without loss of generality that $y = e_N$. By assumption, $\text{dist}(e_N, K) = 1$. We need to show that K is contained in the (closed) lower half-plane.

Assume the contrary: let $x \in K$ be such that $x_N > 0$. Since $0 \in K$ and since K is convex, it follows that $\epsilon x \in K$ for every $\epsilon \in (0, 1)$. Thus,

$$\text{dist}(e_N, K) \leq |e_N - \epsilon x|, \quad \epsilon \in (0, 1).$$

Since $\epsilon \in (0, 1)$ is arbitrary, it follows that

$$\text{dist}(e_N, K) \leq \inf_{\epsilon \in (0, 1)} |e_N - \epsilon x| = \inf_{\epsilon \in (0, 1)} \left(\epsilon^2 \sum_{k=1}^{N-1} x_k^2 + (1 - \epsilon x_N)^2 \right)^{\frac{1}{2}}.$$

Since $x_N > 0$, it follows that infimum is strictly less than 1. Thus, $\text{dist}(e_N, K) < 1$. However, it is granted in the lemma that $\text{dist}(e_N, K) = 1$. This contradiction shows that such an x cannot exist. This completes the proof. $\square$

Lemma 3.6. *If $p \in (N, 2N)$, then*

$$\int_{\Gamma} \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq c_p \text{dist}(gy, \Gamma)^{-N}, \quad y \in \Gamma.$$

Proof. Find a point $z \in \Gamma$ such that

$$|gy - z| = \text{dist}(gy, \Gamma).$$

By Lemma 3.5, affine hyperplane passing through z and orthogonal to $gy - z$ splits the space into two half-spaces, one of them containing gy , the other one containing Γ (denote this one by H).

We have $y \in H$, $\Gamma \subset H$ and $gy \notin H$. So,

$$\int_{\Gamma} \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq \int_H \frac{|x - y|^{p-2N} dx}{|x - gy|^p}.$$

Using Lemma 3.4, we write

$$\int_H \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq c_{p,N} \text{dist}(gy, H)^{-N}.$$

Since $z \in \partial H$ and since $gy - z$ is orthogonal to ∂H , it follows that $\text{dist}(gy, H) = |gy - z| = \text{dist}(gy, \Gamma)$. $\square$

Proof of Theorem 3.1. Case 1: Let $p \in (N, 2N)$.

We have

$$\begin{aligned} & \left(\int_{\Gamma \times g\Gamma} \left(\frac{d(x, y)}{|x - y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x, y)^{2N}} \right)^{\frac{1}{p}} = \\ & = \left(\int_{\Gamma \times \Gamma} \left(\frac{|x - y|}{|x - gy|} \right)^p |f(x) - f(gy)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \left(\int_{\Gamma \times \Gamma} \left(\frac{|x - y|}{|x - gy|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} + \\ & + \left(\int_{\Gamma \times \Gamma} \left(\frac{|x - y|}{|x - gy|} \right)^p |f(y) - f(gy)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \left(\int_{\Gamma \times \Gamma} |f(x) - f(y)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} + \end{aligned}$$

$$+ \left(\int_{\Gamma} |f(y) - f(gy)|^p \left(\int_{\Gamma} \frac{|x-y|^{p-2N}}{|x-gy|^p} dx \right) dy \right)^{\frac{1}{p}}.$$

Obviously, the first summand on the right hand side is less than $\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$. By Lemmas 3.6 and 3.2,

$$\begin{aligned} & \left(\int_{\Gamma \times g\Gamma} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)} + \\ & + c_{p,R}^{(1)} \left(\int_{\Gamma} |f(y) - f(gy)|^p \left(\int_{\Gamma} \frac{dx}{|x-gy|^{2N}} \right) dy \right)^{\frac{1}{p}} = \\ & = \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)} + c_{p,R}^{\frac{1}{p}} \left(\int_{\Gamma \times \Gamma} \frac{|f(y) - f(gy)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)} + c_{p,R}^{(1)} \left(\int_{\Gamma \times \Gamma} \frac{|f(x) - f(gy)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} + \\ & + c_{p,R}^{(1)} \left(\int_{\Gamma \times \Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}}. \end{aligned}$$

Now,

$$\left(\int_{\Gamma \times \Gamma} \frac{|f(x) - f(gy)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} = \left(\int_{\Gamma \times g\Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-y|^{2N}} \right)^{\frac{1}{p}} \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$$

and

$$\left(\int_{\Gamma \times \Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} \leq \left(\int_{\Gamma \times \Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-y|^{2N}} \right)^{\frac{1}{p}} \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Thus,

$$\left(\int_{\Gamma \times g\Gamma} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq (1 + 2c_{p,R}^{(1)}) \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Summing over Γ and $1 \neq g \in G$, we obtain

$$\left(\int_{\mathbb{R}^N \times \mathbb{R}^N} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq |G|^2 (1 + 2c_{p,R}^{(1)}) \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Case 2: Let $p \geq 2N$. We have

$$\begin{aligned} & \left(\int_{\Gamma \times g\Gamma} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \left(\int_{\Gamma \times g\Gamma} \left(\frac{d(x,y)}{|x-y|} \right)^{2N} |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} = \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}. \end{aligned}$$

□

4. PROOF OF THEOREM 1.1 (i)

Let (X, μ) be a measure space and let $L_p(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ denote the mixed-norm space consisting of all $\mu \times \mu$ -measurable functions $f : X \times X \rightarrow \mathbb{C}$ such that

$$\int_X \|f(x, \cdot)\|_{L_{q,\infty}(X,\mu)}^p d\mu(x), \int_X \|f(\cdot, y)\|_{L_{q,\infty}(X,\mu)}^p d\mu(y) < \infty.$$

We equip $L_p(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ with its natural norm.

Lemma 4.1. *Let (X, μ) be a measure space and let $2 < p < \infty$. Suppose that $K \in L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)$. The integral operator*

$$T_K f(x) := \int_X K(x, y) f(y) d\mu(y)$$

belongs to $S_{p,\infty}(L_2(X, d\mu))$. Moreover, we have

$$\|T_K\|_{S_{p,\infty}(L_2(X, d\mu))} \leq c_p \|K\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)}.$$

Proof. The proof is given in [8, Lemma 2] (see also [7, Corollary 11.2]). $\square$

Lemma 4.2. *Let $p > 2$ and let $\frac{1}{q} = 1 - \frac{2}{p}$. We have*

$$\|F_1 F_2\|_{L_p(L_{p',\infty})(X \times X, \mu \times \mu)} \leq 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X \times X, \mu \times \mu)} \|F_2\|_{L_\infty(L_{q,\infty})(X \times X, \mu \times \mu)}.$$

Proof. We clearly have

$$LHS = \left(\int_X \|F_1(x, \cdot) F_2(x, \cdot)\|_{L_{p',\infty}(X,\mu)}^p d\mu(x) \right)^{\frac{1}{p}}.$$

Clearly, $\frac{1}{p'} = \frac{1}{p} + \frac{1}{q}$. Thus,

$$\begin{aligned} \|F_1(x, \cdot) F_2(x, \cdot)\|_{L_{p',\infty}(X,\mu)} &\leq 2^{\frac{1}{p'}} \|F_1(x, \cdot)\|_{L_{p,\infty}(X,\mu)} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)} \\ &\leq 2^{\frac{1}{p'}} \|F_1(x, \cdot)\|_{L_p(X,\mu)} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)}. \end{aligned}$$

Hence,

$$\begin{aligned} LHS &\leq 2^{\frac{1}{pp'}} \left(\int_X \|F_1(x, \cdot)\|_{L_p(X,\mu)}^p \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)}^p d\mu(x) \right)^{\frac{1}{p}} \leq \\ &\leq 2^{\frac{1}{pp'}} \left(\int_X \|F_1(x, \cdot)\|_{L_p(X,\mu)}^p d\mu(x) \right)^{\frac{1}{p}} \times \sup_{x \in X} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)} = \\ &= 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X \times X, \mu \times \mu)} \|F_2\|_{L_\infty(L_{q,\infty})(X \times X, \mu \times \mu)}. \end{aligned}$$

$\square$

Lemma 4.3. *Let (X, μ) be a measure space and let $2 < p < \infty$. Suppose that $F \in L_p(X \times X, H^2(\mu \times \mu))$ and $H \in L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$. We have*

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X, d\mu))} \leq c_p \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \|H\|_{L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}^{1 - \frac{2}{p}}.$$

Proof. By Lemma 4.1,

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X, d\mu))} \leq c_p \|FH\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)}.$$

Let $\frac{1}{q} = 1 - \frac{2}{p}$. It follows from Lemma 4.2 that

$$\|F_1 F_2\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)} \leq$$

$$\leq 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X \times X, \mu \times \mu)} \|F_2\|_{L_\infty(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}.$$

Setting $F_1 = FH^{\frac{2}{p}}$ and $F_2 = H^{\frac{1}{q}}$, we obtain

$$\begin{aligned} & \|FH\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)} \leq \\ & \leq 2^{\frac{1}{pp'}} \|FH^{\frac{2}{p}}\|_{L_p(X \times X, \mu \times \mu)} \|H^{\frac{1}{q}}\|_{L_\infty(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)} = \\ & = 2^{\frac{1}{pp'}} \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \|H\|_{L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}^{\frac{1}{q}} = \\ & = 2^{\frac{1}{pp'}} \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \|H\|_{L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}^{1-\frac{2}{p}}. \end{aligned}$$

□

Lemma 4.4. *Let (X, μ) be a measure space and let $2 < p < \infty$. Suppose that $F \in L_p(X \times X, GH^2(\mu \times \mu))$ and $H \in L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$. We have*

$$\|TFH\|_{S_p(L_2(X, d\mu))} \leq c_{p,H} \|F\|_{L_p(X \times X, H^2(\mu \times \mu))}.$$

Proof. The assertion follows from Lemma 4.3 by real interpolation. □

Proof of Theorem 1.1 (i). In what follows, let $X = \mathbb{R}^N$ and $d\mu(x) = w(x)dx$. Set

$$F(x, y) = w(B(x, d(x, y))) K_{\text{Dunkl},j}(x, y) (f(x) - f(y)), \quad x \neq y,$$

$$H(x, y) = \frac{1}{w(B(x, d(x, y)))}, \quad x \neq y.$$

By [7, Lemma 11.7], $H \in L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$. By Lemma 2.2,

$$\begin{aligned} & \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \leq \\ & \leq c_{R,k}^{(1)} \left(\int_{\mathbb{R}^N \times \mathbb{R}^N} \left(\frac{d(x, y)}{|x - y|} \right)^p |f(x) - f(y)|^p \frac{w(x)w(y)dx dy}{w(B(x, d(x, y)))^2} \right)^{\frac{1}{p}}. \end{aligned}$$

Using Lemma 2.3, we conclude that

$$\|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \leq c_{R,k}^{(2)} \left(\int_{\mathbb{R}^N \times \mathbb{R}^N} \left(\frac{d(x, y)}{|x - y|} \right)^p |f(x) - f(y)|^p \frac{dx dy}{d(x, y)^{2N}} \right)^{\frac{1}{p}}.$$

Using Theorem 3.1, we conclude that

$$\|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \leq c_{R,k} \|f\|_{\dot{W}^{\frac{N}{p}, p}(\mathbb{R}^N)}.$$

Obviously, the integral kernel of the operator $[R_{\text{Dunkl},j}, M_f]$ is FH . The assertion follows now from Lemma 4.4. □

5. PROOF OF THEOREM 1.1 (ii)

Lemma 5.1. *Let $Q \in \mathcal{D}$. We have*

$$K_j(x, y) \geq \frac{c_k}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

Proof. TO BE DONE. Follows from Lemma 2.4. □

Lemma 5.2. *Let real-valued $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ be such that the commutator $[R_{\text{Dunkl},j}, M_f]$ is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. For every $Q \in \mathcal{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$ and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq c_k |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Proof. Let α be the median of f on $3Q$ and β be the median of f on $Q + 3s(Q)e_j$. Assume for definiteness $\alpha \geq \beta$. Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, & B_1 &= \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, & B_2 &= \{x \in Q + 3s(Q)e_j : f(x) \geq \beta\}, \\ \xi_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{A_s}, & \eta_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2. \end{aligned}$$

We have

$$\begin{aligned} &\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ &= \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x, y) (f(y) - f(x)) w(x) w(y) dx dy. \end{aligned}$$

If $x \in B_1$ and $y \in A_1$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ &= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If $x \in B_2$ and $y \in A_2$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \beta) + (\beta - f(x)) = \\ &= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0. \end{aligned}$$

Thus,

$$\begin{aligned} &\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x, y) (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot \left(w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \right) \geq \\ &= \frac{c_k}{w^2(Q)} \cdot \left(\frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \right) \geq \\ &\geq \frac{c_k}{4w(Q)} \cdot \left(\int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \right) \end{aligned}$$

and

$$\begin{aligned} &-\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x, y) |f(y) - \beta| w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq \\ &= \frac{c_k}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} &\int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy + \\ &+ \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq \end{aligned}$$

$$\begin{aligned}
&\leq \int_{A_1} |f(y) - \alpha|w(y)dy + \int_{A_2} |f(y) - \beta|w(y)dy \\
&+ \int_{A_2} |\alpha - \beta|w(y)dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta|w(y)dy \leq \\
&\leq \int_{A_1} |f(y) - \alpha|w(y)dy + \int_{A_2} |f(y) - \beta|w(y)dy + 2|\alpha - \beta|w(3Q).
\end{aligned}$$

Combining the last 3 equations, we conclude

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha|w(y)dy \leq \max_{s=1,2} |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$$|\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta_Q = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha|w(y)dy \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Hence,

$$c'_k |f_{3Q} - \alpha| \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|$$

and, finally,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - f_{3Q}|w(y)dy \leq 2|\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

□

Lemma 5.3. *Suppose that $1 < p < \infty$ and $f \in L_1^{\text{loc}}(\mathbb{R}^N)$. If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ for some $j \in \{1, 2, \dots, N\}$, then*

$$\begin{aligned}
&\left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}|w(x)dx \right\}_{Q \in \mathcal{D}} \right\|_{l_p(\mathcal{D})} \leq \\
&\leq c_k \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}.
\end{aligned}$$

If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ for some $j \in \{1, 2, \dots, N\}$, then

$$\begin{aligned}
&\left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}|w(x)dx \right\}_{Q \in \mathcal{D}} \right\|_{l_{N,\infty}(\mathcal{D})} \leq \\
&\leq c_k \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}.
\end{aligned}$$

Proof. We prove only the first assertion. The proof of the second one follows *mutatis mutandi*.

Let $\{\xi_Q\}_{Q \in \mathcal{D}}$ and $\{\eta_Q\}_{Q \in \mathcal{D}}$ be functions constructed in Lemma 5.2. By Lemma 2.6,

$$\left\{ \langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q \in \mathcal{D}} \in l_p.$$

The first assertion follows now from Lemma 5.2. □

Lemma 5.4. *Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x \in Q} w(x) \leq \frac{c_k}{m(Q)} \int_Q w(x)dx.$$

Proof. If $x = c_Q + y$, $y \in [-\frac{1}{2}s(Q), \frac{1}{2}s(Q)]^d$, then

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + r|\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2}s(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2}s(Q))^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2}s(Q))^{k(v)}.$$

Fix chamber Γ such that $c_Q \in \Gamma$. We have

$$\int_Q w(x)dx \geq \int_{c_Q + (B(0, \frac{1}{2}s(Q)) \cap \Gamma)} w(x)dx = \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} w(c_Q + y)dy.$$

If $y \in \Gamma$, then

$$|\langle c_Q + y, v \rangle| = |\langle c_Q, v \rangle| + |\langle y, v \rangle|, \quad v \in R.$$

Thus,

$$w(c_Q + y) = \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x)dx \geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)} dy.$$

Fix a closed cone $\Gamma' \subset \Gamma$ such that (i) interior of Γ' is non-empty and (ii) $\partial\Gamma \cap \partial\Gamma' = \{0\}$. We have

$$|\langle y, v \rangle| \geq c_{\Gamma'}|y|, \quad y \in \Gamma', \quad v \in R.$$

We, therefore, have

$$\begin{aligned} \int_Q w(x)dx &\geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma'} \prod_{v \in R} (|\langle c_Q, v \rangle| + c_{\Gamma'}|y|)^{k(v)} dy \geq \\ &\geq \int_{\frac{1}{4}s(Q) \leq |y| \leq \frac{1}{2}s(Q)} \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{2}c_{\Gamma'}s(Q))^{k(v)} dy = \\ &= s(Q)^N \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{4}c_{\Gamma'}s(Q))^{k(v)} \times (2^{-N} - 4^{-N}) \int_{\substack{y \in \Gamma' \\ |y| \leq 1}} dy. \end{aligned}$$

Combining this with the preceding paragraph, we complete the proof. $\square$

Lemma 5.5. Suppose $1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}$. Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have

$$\left(\frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx} \right)^{\frac{1}{t}} \leq c_{t,k} \frac{m(Q)}{\int_Q w(x)dx}.$$

Proof. If $v \in R$ is such that

$$\sqrt{d}s(Q) \leq |\langle c_Q, v \rangle|,$$

then

$$|\langle x, v \rangle| \geq |\langle c_Q, v \rangle| - |\langle x - c_Q, v \rangle| \geq |\langle c_Q, v \rangle| - \frac{\sqrt{d}}{2}s(Q) \geq \frac{1}{2}|\langle c_Q, v \rangle|, \quad x \in Q.$$

In this case,

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq 2^{(t-1)|R|k(v)} |\langle c_Q, v \rangle|^{(1-t)|R|k(v)} \cdot \int_Q dx = \\ &= 2^{(t-1)|R|k(v)} |\langle x_0, v \rangle|^{(1-t)|R|k(v)} \cdot s(Q)^N. \end{aligned}$$

If $v \in R$ is such that

$$|\langle c_Q, v \rangle| \leq \sqrt{d}s(Q),$$

then let $y_0 = \langle c_Q, v \rangle$ and $z_0 = c_Q - y_0 v$. Let us introduce new variables $y = \langle x, v \rangle$ and $z = x - yv$ (it varies in the plane orthogonal to v). We write

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq \int_{|x-c_Q| \leq \frac{\sqrt{d}}{2}s(Q)} |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq \\ &\leq \int_{|y-y_0|, |z-z_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy dz = \\ &= \left(\frac{\sqrt{d}}{2}s(Q)\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y-y_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy \leq \\ &= \left(\frac{\sqrt{d}}{2}s(Q)\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy = \\ &= s(Q)^{N+(1-t)|R|k(v)} \cdot \left(\frac{\sqrt{d}}{2}\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy. \end{aligned}$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq c_1 s(Q)^N \left(\max\{|\langle c_Q, v \rangle|, r\} \right)^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\begin{aligned} \int_Q w^{1-t}(x) dx &\leq \prod_{v \in R} \left(\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \right)^{\frac{1}{|R|}} \leq \\ &\leq c_2 s(Q)^N \prod_{v \in R} \left(\max\{|\langle c_Q, v \rangle|, s(Q)\} \right)^{(1-t)k(v)}. \end{aligned}$$

The assertion follows easily. $\square$

Lemma 5.6. *Measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [6].*

Proof. That both measures are doubling is obvious.

Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). By Lemmas 5.4 and 5.5,

$$\begin{aligned} \left(\frac{1}{m(Q)} \int_Q w^t(x) dx \right)^{\frac{1}{t}} &\leq \frac{c_k}{m(Q)} \int_Q w(x) dx, \quad t > 0. \\ \left(\frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx} \right)^{\frac{1}{t}} &\leq c_G \frac{m(Q)}{\int_Q w(x) dx}, \quad 1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}. \end{aligned}$$

The assertion follows now from Proposition 3.6 in [6]. $\square$

Proof of Theorem 1.1 (ii). The left hand side in Lemma 5.3 is the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. So, the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

Lemma 5.6 asserts that measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [6]. Appealing to Proposition 1.2 in [6], we conclude that the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the one with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. By the preceding paragraph, the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

That oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is equivalent to the Sobolev semi-norm of the function f is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [7]. The assertion follows now from the preceding paragraph. $\square$

Proof of Theorem 1.1 (iv). The argument follows the one in the proof of Theorem 1.1 (ii) *mutatis mutandi*. $\square$

6. CONSTANT CHARACTERIZATION

Theorem 6.1. *If $f \in L_1^{\text{loc}}(\mathbb{R}^N, w(x)dx)$ is such that*

$$[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$$

for some $p \leq d$, then f is constant.

Proof. By Lemma 5.6, measures dx and $w(x)dx$ and operator $T_{\Gamma,j} = M_{\chi_\Gamma} R_{\text{Dunkl},j} M_{\chi_\Gamma}$ satisfy **on Γ** the assumptions in Theorem 1.4 in [6]. By assumption, $[T_{\Gamma,j}, M_f] \in \mathcal{L}_p(L_2(\Gamma, w(x)dx))$. By Theorem 1.4 in [6], f is constant on Γ .

Hence, f is constant on every chamber. Let $f = c_1$ on Γ_1 and $f = c_2$ on Γ_2 . On the other hand, $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{2N}(L_2(\mathbb{R}^N, w(x)dx))$. By Theorem 1.1 (ii), $f \in \dot{W}^{\frac{1}{2}, 2N}(\mathbb{R}^N)$. We have

$$\begin{aligned} \int_{\Gamma_1 \times \Gamma_2} \frac{|c_1 - c_2|^{2N} dx dy}{|x - y|^{2N}} &= \int_{\Gamma_1 \times \Gamma_2} \frac{|f(x) - f(y)|^{2N} dx dy}{|x - y|^{2N}} \leq \\ &\leq \int_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|f(x) - f(y)|^{2N} dx dy}{|x - y|^{2N}} = \|f\|_{\dot{W}^{\frac{1}{2}, 2N}(\mathbb{R}^N)}^{2N} < \infty. \end{aligned}$$

It follows that $c_1 = c_2$. Since chambers Γ_1 and Γ_2 are arbitrary, the assertion follows. $\square$

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REFERENCES

- [1] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. [5](#)
- [2] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. [5](#)
- [3] Z. Fan, M.T. Lacey, J. Li, and X. Xiong. Schatten–Lorentz characterization of Riesz transform commutator associated with Bessel operators. Preprint available at <https://arxiv.org/abs/2403.08249>, 2024. [5](#)
- [4] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. [3](#), [4](#), [5](#)
- [5] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. [3](#)
- [6] T. Hytonen. A_∞ -invariance of oscillatory norms and schatten characterisations of commutators. [2](#), [15](#), [16](#)
- [7] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at <https://arxiv.org/abs/2411.02613>, 2024. [5](#), [10](#), [11](#), [16](#)
- [8] S. Janson and T.H. Wolff. Schatten classes and commutators of singular integral operators. *Ark. Mat.*, 20(2):301–310, 1982. [10](#)
- [9] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. [5](#)
- [10] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. [5](#)
- [11] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon-Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. [5](#)
- [12] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at <https://arxiv.org/abs/2411.05810>, 2024. [5](#)
- [13] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at <https://arxiv.org/abs/2407.14988>, 2024. [5](#)

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